

Demonstrating Talking Spell: A Wearable System for Real-Time Anthropomorphic Interaction with Everyday Objects

Xuetong Wang
Division of Emerging
Interdisciplinary Areas
Hong Kong University of Science and
Technology
Kowloon, Hong Kong SAR, China
xwangdd@connect.ust.hk

Ching Christie Pang
Division of Emerging
Interdisciplinary Areas
Hong Kong University of Science and
Technology
Hong Kong SAR, China
ccpangaa@connect.ust.hk

Pan Hui^{*†}
Computational Media and Arts
The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, Guangdong, China
panhui@ust.hk



Figure 1: Talking Spell consists the scope (camera for detecting objects) and wand (sensor to allow spell casting and vibration motor to inform useful "spell". Example scope devices we created include (colored in green, clockwise from top-left): head chain, glasses, and bone conduction headphones; example wand devices we created include (colored in red, clockwise from top-left): a necklace with sensor pendant, bag charm, earrings, and a part of bone conduction headphones. Note that the two components (top-right) form an all-in-one device in the bone conduction headphones.

Abstract

Virtual assistants, common in devices like smartphones, often struggle to form deep emotional connections due to their fixed forms. Talking Spell, a novel wearable system, addresses this by enabling users to endow everyday objects with speech and anthropomorphic personas, fostering engaging human-object interactions. Leveraging advanced computer vision (e.g., YOLOv11 for object detection),

large vision-language models (e.g., QWEN-VL for persona generation), speech-to-text and text-to-speech technologies, Talking Spell guides users through emotional bonding stages, enhancing experiences in entertainment, companionship, utility, and creativity. We aim to advance AI companionship, with potential applications in education, mental health, and accessibility. Future developments will focus on integrating Talking Spell into smart environments and creating adaptive personas that evolve, deepening user connections.

^{*}Corresponding author

[†]Pan Hui is also affiliated with Hong Kong University of Science and Technology, and University of Helsinki, Finland

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758985>

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design; Ubiquitous and mobile computing systems and tools.**

Keywords

Embodied and Explorable Interaction; AI Companionship; Wearable; On-body Devices; Ubiquitous Computing; Human-Object Interaction; Large Language Models (LLMs)

ACM Reference Format:

Xueting Wang, Ching Christie Pang, and Pan Hui. 2025. Demonstrating Talking Spell: A Wearable System for Real-Time Anthropomorphic Interaction with Everyday Objects. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758985>

1 Introduction

The rapid advancement of artificial intelligence (AI) has significantly enhanced human-computer interaction (HCI), particularly through virtual assistants (VAs) such as Alexa and Siri, which are now integral to smartphones, home devices, and wearables [1, 7]. Despite their widespread adoption, these assistants often function as standalone entities, limiting their ability to facilitate rich, context-aware interactions with users' environments [8]. Prior research has primarily focused on the functional capabilities of VAs, often overlooking opportunities for deeper, more personalized engagement with everyday objects due to a lack of contextual understanding [2, 4, 5, 11]. This limitation confines interactions to a singular, rigid model, preventing users from fully realizing the potential of AI companionship.

To address this challenge, we propose a context-aware AI interaction framework that leverages anthropomorphism—the attribution of human-like characteristics to non-human entities—to strengthen emotional and cognitive connections between users and objects [3, 6, 9]. Inspired by creative depictions such as the Talking Spell in Harry Potter, which imbues objects with the ability to communicate, we introduce Talking Spell, a novel system designed to enable dynamic, meaningful interactions with everyday objects. This system employs hardware components, including a scope for object detection and a wand for issuing voice-activated commands, allowing users to "cast spells" and engage directly with their surroundings [1]. By infusing anthropomorphic features, Talking Spell adapts its functionality to the user's context, fostering a seamless and engaging interaction paradigm. Our contributions include a communication pipeline that enhances user engagement through anthropomorphism, an exploration of the design space for human-object interaction tools, and the creation of new possibilities for creativity and contextual awareness [4, 10]. Ultimately, Talking Spell aims to transform AI companionship into a versatile, adaptive tool that enriches user experiences across both practical and imaginative scenarios, redefining how users connect with their environments.

2 Design and Implementation

We present the hardware architecture and implementation of Talking Spell, a wearable system comprising two core components: the scope and the wand. The scope serves as the system's visual processing unit, functioning as an observational "eye" that captures images and performs object detection. In contrast, the wand acts as an interaction trigger, activating the dialogue system powered by a

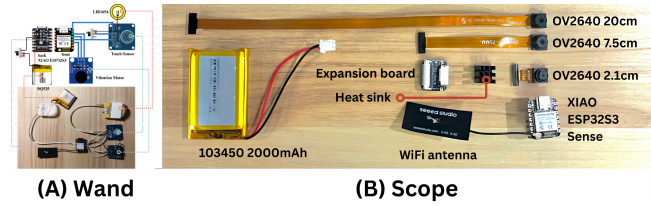


Figure 2: The design and components of scope (left) for triggering the interaction; and wand (right) for the object recognition.

large language model (LLM). This section provides a detailed examination of these components, followed by an analysis of the design considerations that ensure portability and seamless integration into users' daily lives.

2.1 Scope

The scope serves as the visual component of the Talking Spell system, designed to capture images and perform object detection. As illustrated in Figure 2, the scope employs the XIAO ESP32S3 Sense as its primary processing board, integrated with the OV2640 camera module. This configuration allows the scope to stream video via a web server to a designated IP address by establishing a connection to a computer-generated hotspot. Users can access real-time image frames by visiting this IP address on a local computer.

To ensure high image quality, the scope must be positioned at eye level, allowing the captured images to align closely with the user's visual perspective. We selected three different sizes of the OV2640 camera module—2.1 cm, 7.5 cm, and 20 cm—to accommodate various wearable needs. This flexibility in size ensures that the scope can be integrated into different forms of wearable technology, enhancing user experience while maintaining functionality.

2.2 Wand

The wand functions as the activation device for the Talking Spell system, facilitating user interaction with the environment. As depicted in Figure 2, the wand is also built around the XIAO ESP32S3 as its primary processing board. The design integrates additional components, including two toggle switches, a touch sensor module (TTP233), a vibration motor module, and two 3.7V batteries (LIR1654, 502525 3.7V 300mAh Li-Ion Polymer Battery), which independently power the various components.

Upon detecting a signal from the touch sensor, the main board transmits this information via Bluetooth to a connected computer, initiating a dialogue request with the LLM. The system then receives a recording initiation signal from the computer, which activates the vibration motor module. This sequence creates an intuitive user interaction workflow: pressing the touch sensor starts the recording, the vibration motor provides haptic feedback, and releasing the sensor stops the recording process. This design allows for a seamless interaction experience, ensuring that users can easily communicate with the system.

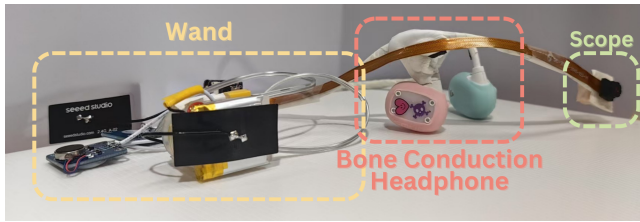


Figure 3: A bone conduction headphone as an example of Talking Spell integrated into an all-in-one wearable device.

2.3 Implementation and Our Portable Design

Our design fully embodies the pursuit of portability, ensuring that the hardware of Talking Spell seamlessly integrates into users' daily lives. We made careful trade-offs in size to achieve this goal. The XIAO ESP32S3 Sense was selected as the visual module due to its compact dimensions of only 21 mm x 18 mm, making it smaller than other ESP-CAM modules. Instead of relying on the USB-C port for power, we utilized the external removable 103450 Lithium-ion polymer rechargeable battery located at the back of the device. Additionally, we reduced the image size to 320x320 pixels to optimize image transmission quality via Wi-Fi.

In the design of the wand, we initially considered using Wi-Fi for signal transmission. However, testing revealed that the batteries meeting our size requirements could not provide sufficient current to activate the Wi-Fi module effectively. Consequently, we opted for Bluetooth for low-power transmission. Given the small size and capacity of the chosen batteries, they could provide only a limited discharge current. To prevent functional loss due to current sharing among modules, we implemented separate power supplies for the touch module and the main board, ensuring the proper operation of each component.

After multiple iterations of design and testing, we ultimately achieved a compact design size of 35 mm x 25 mm x 25 mm for the wand, balancing functionality and portability. This careful consideration of design elements ensures that Talking Spell can be easily integrated into the daily lives of users, enhancing their interaction with everyday objects while maintaining an unobtrusive presence. To make Talking Spell portable, we designed various wearables so that it can be easily carried around, handy, and fashionable, as shown in 1. 3 is the example of all-in-one wearable that embeds both scope and wand.

3 Conclusion

Talking Spell presents an innovative wearable system that facilitates real-time anthropomorphic voice interactions with everyday objects through the integration of advanced artificial intelligence and compact hardware. By employing YOLOv11 for object detection, QWEN-VL for persona generation, and various speech technologies, Talking Spell effectively addresses the limitations of existing interaction paradigms, fostering personalized user experiences. The implementation and demonstration of portable applications underscore the feasibility and potential of Talking Spell for migratable and personalized interactions across diverse scenarios. This system

paves the way for transformative advancements in AI companionship, with promising applications in education, mental health, and accessibility, fundamentally reshaping how users engage with their environment.

Acknowledgments

This work is partially supported by the National Key Research and Development Program of China under Grant No.: 2024YFC3307602, and the Guangdong Provincial Talent Program under Grant No.: 2023JC10X009.

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